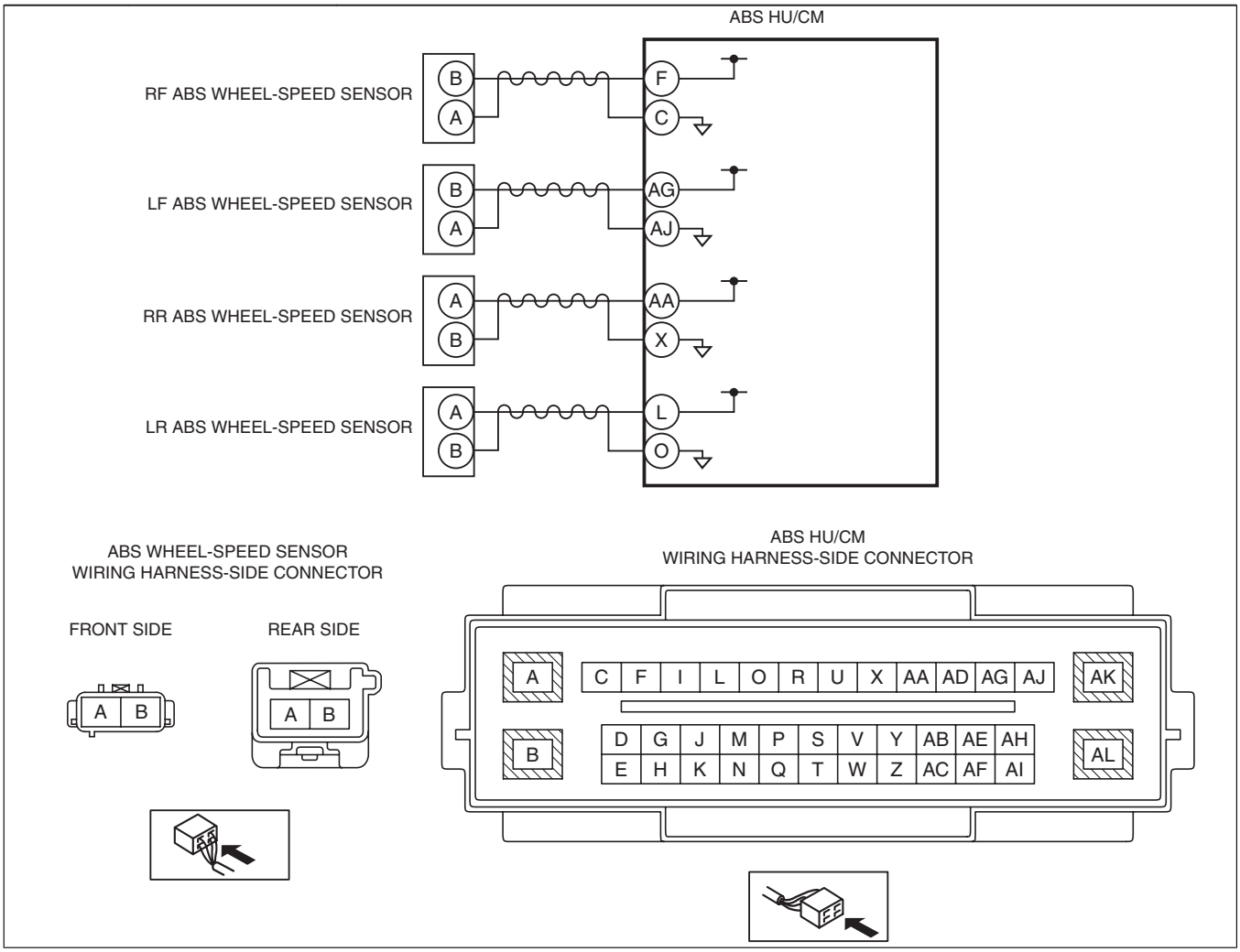
DTC C0030:07/C0031:07/C0031:29/C0031:2F/C0031:64/C0033:07/C0034:07/C0034:29/C0034:2F/C0034:64/C0036:07/C0037:07/C0037:29/C0037:2F/C0037:64/C0039:07/C003A:07/C003A:29/C003A:2F/C003A:64 [ABS HU/CM]

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DTC	**C0030:07**	**LF ABS sensor rotor**
	C0031:07	**LF ABS wheel-speed sensor/ABS sensor rotor**
	C0031:29	
	C0031:2F	
	C0031:64	
	C0033:07	**RF ABS sensor rotor**
	C0034:07	**RF ABS wheel-speed sensor/ABS sensor rotor**
	C0034:29	
	C0034:2F	
	C0034:64	
	C0036:07	**LR ABS sensor rotor**
	C0037:07	**LR ABS wheel-speed sensor/ABS sensor rotor**
	C0037:29	
	C0037:2F	
	C0037:64	
	C0039:07	**RR ABS sensor rotor**
	C003A:07	**RR ABS wheel-speed sensor/ABS sensor rotor**
	C003A:29	
	C003A:2F	
	C003A:64	
DETECTION CONDITION		• C0030:07, C0033:07, C0036:07, C0039:07 — Periodic abnormality is detected in the signal wave pattern from the ABS wheel-speed sensors • C0031:07, C0034:07, C0037:07, C003A:07 — A large, sudden change in the wheel speed-signal is detected • C0031:29, C0034:29, C0037:29, C003A:29 — While the vehicle is traveling at a speed of 10 km/h {6.2 mph} or more, no signal in any of the four wheels or an extremely low vehicle speed signal is detected • C0031:2F, C0034:2F, C0037:2F, C003A:2F — The wheel speed or acceleration speed in any of the 4 wheels is not within the specification — ABS control continues to operate for 28 s or more • C0031:64, C0034:64, C0037:64, C003A:64 — While the vehicle is traveling at a speed of 20 km/h {12 mph} or more, an extremely high vehicle speed signal in any of the four wheels is detected
FAIL-SAFE FUNCTION		Refer to DTC TABLE. (See DTC TABLE [ABS HU/CM].)
POSSIBLE CAUSE		• Short to GND in wiring harness between ABS wheel-speed sensor and ABS HU/CM • Excessive clearance between the ABS wheel-speed sensor and sensor rotor • ABS wheel-speed sensor malfunction • ABS sensor rotor malfunction (missing ABS sensor rotor teeth due to foreign material obstruction) • ABS sensor rotor installation malfunction (If the ABS sensor rotor is installed at an angle, it may cause output of abnormal wave pattern at high speeds) • Continuous ABS operation



Diagnostic Procedure

Step	Inspection		Action
1	**INSPECT ABS WHEEL-SPEED SENSOR OUTPUT** • Using the M-MDS, perform the PID inspection as followings: (See PID/DATA MONITOR INSPECTION [ABS HU/CM].) — WSPD_SEN_LF — WSPD_SEN_LR — WSPD_SEN_RF — WSPD_SEN_RR • Drive the vehicle. • Are the vehicle speeds approximately the same?	Yes	Go to Step3.
		No	Go to the next step.

Step	Inspection		Action
2	**INSPECT ABS WHEEL-SPEED SENSOR SIGNAL CIRCUIT FOR SHORT TO GROUND** • Disconnect the ABS wheel-speed sensor connector. • Inspect for continuity as followings: — RF ABS wheel-speed sensor terminal A (wiring harness-side)—body ground (SKYACTIV-G 1.3, SKYACTIV-G 1.5) — LF ABS wheel-speed sensor terminal A (wiring harness-side)—body ground (SKYACTIV-G 1.3, SKYACTIV-G 1.5) — RR ABS wheel-speed sensor terminal A (wiring harness-side)—body ground (SKYACTIV-G 1.3, SKYACTIV-G 1.5) — LR ABS wheel-speed sensor terminal A (wiring harness-side)—body ground (SKYACTIV-G 1.3, SKYACTIV-G 1.5) • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector in the followings: • RF ABS wheel-speed sensor terminal A—ABS HU/CM terminal C • LF ABS wheel-speed sensor terminal A—ABS HU/CM terminal AJ • RR ABS wheel-speed sensor terminal B—ABS HU/CM terminal X • LR ABS wheel-speed sensor terminal B—ABS HU/CM terminal O **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step5.
		No	Go to the next step.
3	**INSPECT CLEARANCE between the ABS wheel-speed sensor and sensor rotor** • Measure the clearance between the ABS wheel-speed sensor and the ABS sensor rotor. (See FRONT ABS WHEEL-SPEED SENSOR INSPECTION.) (See REAR ABS WHEEL-SPEED SENSOR INSPECTION.) • Is the clearance normal?	Yes	Go to the next step.
		No	Replace the malfunction part, then go to Step5.
4	**VISUALLY INSPECT ABS SENSOR ROTOR** • VISUALLY INSPECT the ABS SENSOR ROTOR FOR FOREIGN MATERIAL ADHERING OR IMPROPER INSTALLATION. • Is the ABS SENSOR ROTOR normal?	Yes	Go to the next step.
		No	Replace the ABS sensor rotor, then go to the next step. • Front ABS sensor rotor (front wheel hub) (See WHEEL HUB, STEERING KNUCKLE REMOVAL/INSTALLATION.) • Rear ABS sensor rotor (rear wheel hub component) (See WHEEL HUB COMPONENT REMOVAL/INSTALLATION [REAR BRAKE (DISC)].)
5	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Using the M-MDS, clear the DTC from the ABS HU/CM. (See CLEARING DTC [ABS HU/CM].) • Start the engine and drive the vehicle at 10 km/h {6.2 mph} or more. • Using the M-MDS, perform the ABS HU/CM DTC inspection. (See DTC INSPECTION [ABS HU/CM].) • Is the same DTC present?	Yes	Repeat the inspection from Step1. If the malfunction recurs, replace the ABS HU/CM, then go to the next step. (See ABS HU/CM REMOVAL/INSTALLATION [L.H.D.].) (See ABS HU/CM REMOVAL/INSTALLATION [R.H.D.].)
		No	Go to the next step.
6	**VERIFY NO DTC IS PRESENT** • Are any DTCs present?	Yes	Go to applicable DTC inspection. (See DTC TABLE [ABS HU/CM].)
		No	DTC troubleshooting completed.